

Dylan Pieper

Data Scientist

Personal Information

He/Him
Madison, Wisconsin
dylanpieper@gmail.com
<https://dylanpieper.github.io/>

Core Competencies

Soft Skills

Cross-functional Collaboration,
User Support & Feedback,
Technical Communication,
Data Strategy &
Roadmap Planning,
Interviewing,
Writing Documentation

Data Tools

SQL (MySQL, PostgreSQL, DuckDB),
REDCap, Git, GitHub, Sentry,
Docker, Azure, Microsoft Foundry,
Linux (RHEL), HTML/CSS/JS/PHP,
WordPress, SharePoint

Infrastructure Tools

SQL (MySQL, PostgreSQL, DuckDB),
REDCap, Git, GitHub, Sentry,
Docker, Azure, Microsoft Foundry,
Linux (RHEL), HTML/CSS/JS, PHP,
WordPress, SharePoint

Languages

English

Native

Mandarin

Beginner

Experience

Research Analyst - Collective for Innovation, Scholarship, and Research in Undergraduate Medical Education (CISR-UME), School of Medicine and Public Health, University of Wisconsin-Madison

Nov 2025 - Present

- Process data using LLMs on a Linux Azure VM ("Platform X") and Microsoft Foundry, including:
 - Formative feedback on objective structured clinical examinations (OSCEs) by transcribing video recordings using OpenAI's Whisper model and natural language processing
 - First-pass grading on open-ended responses for course assignments
- Develop automated Quarto reports to communicate research findings and assessment data to medical school faculty and administrative staff
- Build and deploy the Medical Education Research Hub, a Shiny application on Posit Connect serving as a centralized directory of medical education research projects (<https://connect.doit.wisc.edu/meded-research-hub>)
- Write grant proposals for the Wisconsin Partnership Program, Macy Foundation, and National Board of Medical Examiners
- Build cross-unit fluency in operational tools and processes spanning the Medical Education Office and Informatics and Information Technology, mobilizing campus resources to coordinate work across teams

Data Scientist - School of Pharmacy, University of Pittsburgh, Pittsburgh, PA

Sep 2021 - Oct 2025

- Led data strategy for grant projects with IT departments, researchers, technical assistance specialists, healthcare providers, and public officials
- Developed and deployed surveys and forms (Qualtrics, REDCap, Ninja Forms, and SurveyJS), automated reports (Quarto), and dashboards (Shiny and Tableau) to collect data and communicate findings to project teams and partners
- Engineered SQL database linking clinical, operational, and geographic data across projects
- For a project funded by the state, used Bayesian multilevel modeling to predict risk factors for treatment retention across the 250+ treatment center network, designed REDCap integrations of electronic medical records, and built automated ETL workflows in R and DuckDB

Experience

- For a project funded by Arnold Ventures, used frequentist multilevel modeling to predict racial disparities in pretrial bail decisions, adjusting for key covariates, and built interactive D3 Sankey visualizations of case outcomes across courts from arraignment through sentencing
- Contributed to mixed-methods evaluation, integrating qualitative data from key informant interviews (with district attorneys and judges) conducted by evaluators alongside quantitative analysis and psychometrics
- Contributed to training junior data staff, including recruitment, professional development, and mentorship

Field Interviewer - NORC, University of Chicago, Chippewa Falls, Wisconsin

Aug 2021 - Nov 2021

- Conducted structured interviews with research participants across diverse geographic locations, building rapport and ensuring high-quality data collection using standardized survey instruments and protocols
- Managed project coordination and logistics including scheduling, travel planning, and participant communication to ensure successful data collection across assigned geographic areas
- Maintained strict data governance and research ethics standards, ensuring confidentiality and security protocols throughout the participant interactions

Research Specialist - Michele Gelfand's Culture Lab, University of Maryland, College Park, MD

Aug 2020 - Aug 2021

- Analyzed and published data on COVID-19 deaths across 57 countries and their relationship to social norms using multilevel statistical models
- Ran two experiments testing different nudges, or message framing effects, for wearing masks and forecasted the utility of these interventions
- Developed a natural language processing (NLP) Shiny app to detect threatening language and compare levels across texts (<https://www.michelegelfand.com/threat-dictionary>)

Writing Coach - University of Northern Iowa, Cedar Falls, IA

Aug 2017 - May 2020

- Coached students in writing and statistics, providing personalized guidance based on individual needs assessment to improve academic performance across diverse learning styles
- Trained new tutors and developed training materials to ensure consistent, effective tutoring practices and knowledge transfer across the team
- Conducted outreach in classrooms and registration events to promote tutoring services and connect students with academic support resources

Experience

- Applied evidence-based frameworks including Bloom's taxonomy and Csikszentmihalyi's flow state model to optimize student learning experiences
- Supported diverse student populations including English Language Learners and students with learning disabilities, adapting approaches to meet varied accessibility needs

Education

Master of Arts - Social Psychology - University of Northern Iowa, Cedar Falls, Iowa

2018 - 2020

- Completed courses Statistical Computing I and II in R programming
- Worked as a Research Assistant in the Psychoneuroendocrinology Lab
- Worked as a Teaching Assistant in undergraduate psychology courses (Introduction to Psychology, Biopsychology, and Psychology of Gender)
- Awarded research and travel funds from the university totaling \$1,800
- Thesis: "Challenging Social Systems Under the Threat of Pollution" (<https://scholarworks.uni.edu/etd/1028/>)

Bachelor of Arts - Psychology - University of Northern Iowa, Cedar Falls, Iowa

2014 - 2018

Publications

- Gelfand, M. J., Jackson, J. C., Pan, X., Nau, D., **Pieper, D.**, Denison, E., Dagher, M., Van Lange, P. A. M., Chiu C., & Wang, M. (2021). The relationship between cultural tightness-looseness and COVID-19 cases and deaths: A global analysis. *The Lancet: Planetary Health*. [https://doi.org/10.1016/S2542-5196\(20\)30301-6](https://doi.org/10.1016/S2542-5196(20)30301-6)
- Gelfand, M. J., Li, R., Stamkou, E., Denison, E., Fernandez, J., Choi, V., Chatmen, J., Jackson, J. C., **Pieper, D.**, & Dimant, E. (2021). Persuading conservatives and liberals to comply with mask-wearing: An intervention tournament. *Journal of Experimental Social Psychology*. <https://doi.org/10.1016/j.jesp.2022.104299>
- Dimant, E., Clemente, E. G., **Pieper, D.**, & Azevedo, F. (2022). Politicizing mask-wearing: Predicting the success of behavioral interventions among Republicans and Democrats in the U.S. *Scientific Reports*, 12, 7575. <https://doi.org/10.1038/s41598-022-10524-1>
- **Pieper, D.** (2020). Challenging social systems under the threat of pollution: Replication and extension of Eadeh and Chang (2019) [Master's Thesis, University of Northern Iowa]. UNI ScholarWorks. <https://scholarworks.uni.edu/etd/1028>

Presentations

- **Pieper, D.** (2025, September). From messy to meaningful data: LLM-powered classification in R. *posit::conf*. <https://github.com/dylanpieper/posit-25>
- **Pieper, D.** (2025, June). redquack: An R package for memory efficient REDCap-to-DuckDB workflows. *R/Medicine*. <https://r-consortium.org/posts/redquack-an-r-package-for-memory-efficient-redcap-to-duckdb-workflows>
- Katie, W. & **Pieper, D.** (2024, June). The Pennsylvania medications for opioid use disorder (PA MOUD) technical assistance and quality improvement expansion project. *Rural Health Information Hub*. (Online resource). <https://www.ruralhealthinfo.org/project-examples/1139>
- **Pieper, D.**, Gelfand, M. J., & Denison, E., (2021, February). COVID-19: Meta-norms to wear masks and social distance in the U.S. *Society for Personality and Social Psychology*. (Remote poster session). <https://osf.io/y8cug>
- **Pieper, D.**, Hirano, H., & Harton, H. C. (2020, April). The effect of environmental threats and terrorism/violent-crime threats on political orientation and attitudes. *Midwestern Psychological Association Conference*. Chicago, Illinois. (Conference canceled) ... Also (2020) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference*. Cedar Falls, Iowa. (Conference canceled). <https://osf.io/3n7r8>
- **Pieper, D.** (2020, February). Challenging the system under the threat of pollution: Low system justification mediates a liberal political shift. *Society for Personality and Social Psychology*. New Orleans, LA. (Poster session) ... Also (2020) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference*. Cedar Falls, Iowa. (Conference canceled). <https://osf.io/v7nez>
- **Pieper, D.** (2019, April). What's in a name? Measuring the potential influence of medical cannabis labels on attitudes toward its use. *Iowa Psychological Association Conference*, Des Moines, Iowa. (Poster session) ... Also (2019, April) at the *University of Northern Iowa's College of Social and Behavioral Sciences (CSBS) Conference*. Cedar Falls, Iowa. Awarded Best Oral Presentation (2nd place, \$250). <https://scholarworks.uni.edu/agss/2019/all/18>
- **Pieper, D.**, DeSoto, C., & Hitlan, R. (2018, April). The relations between personality and source of dietary protein. *Midwestern Psychological Association Conference*. Chicago, Illinois. (Poster session) ... Also (2017, April) at the *University of Northern Iowa's College of Social and Behavioral Sciences Conference*. Cedar Falls, Iowa.

R Packages

- **redquack**: Transfer REDCap data to a database and use in R without exceeding available memory (<https://dylanpieper.github.io/redquack>)